Model Question Paper

MAHATMA GANDHI UNIVERSITY KOTTAYAM

MGU- UGP (HONOURS) REGULAR EXAMINATION

FIFTH SEMESTER

Skill Enhancement Course (SEC) - MG5SECMAT300
Introduction to Python for Mathematical Computation

2024 Admission Onwards

Maximum Time:-75 Minutes **Maximum Marks:-50**

Part A

(Short Answer Type Questions)
Answer any 5 questions
Each question carries 2 marks

1. Determine the data type of the variable x after each assignment:

$$x = 10, \qquad x = 10.5, \qquad x = 3 + 4j, \qquad x = \text{'Python'}.$$

[U] / [CO1]

2. Consider the following code:

```
x = 1
while x <= 6:
    if x == 3:
        x = x + 1
        continue
    if x == 5:
        break
    print(x)
    x = x + 1
```

What will be the output? Explain the role of `continue` and `break` in the program. **[U] / [CO1]**

3. Write a Python program using a `for` loop and an `if` statement to print all even numbers from 1 to 20 **[U] / [CO1]**

4. What is NumPy? State any two advantages of using NumPy arrays over Python lists. **[A] / [CO3]**

5. What is the purpose of the `arange()` function? Write the output of:

```
arange(2.0, 3.0, 0.2)
```

[A] / [CO3]

6. Define the dot product of two vectors. Using

$$A = [1, 2, 3], \qquad B = [4, 5, 6],$$

find their dot product. **[A] / [CO3]**

7. What is meant by a subplot in Matplotlib? What do the three arguments in `subplot(2,1,1)` represent?

[A] / [CO4]

8. Define a polar plot. Write the Matplotlib statement used to plot a point using polar coordinates (θ, r).
[A] / [CO4]

Part B
(Short Essay Questions)
Answer any 4 questions
Each question carries 5 marks

9. Write a Python program to read numbers from the keyboard until their sum exceeds 200. Modify the program so that numbers greater than 99 are ignored. Explain the working of the program.

[U] / [CO1]

10. Explain how a Python list can be converted into a NumPy array. Write a Python program to create the following two-dimensional array:

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

and display the array. **[A] / [CO3]**

11. Explain the working of the `arange()` and `linspace()` functions with their syntax. Write Python statements to generate:

    (a) an array from 2 to 3 with a step size of 0.2,

    (b) an array of 11 equally spaced values from 1 to 2.

[A] / [CO3]

12. What is a vectorized function? Explain how vectorized functions eliminate the need for explicit loops. Using NumPy, evaluate

$$\log_{10}(x)$$

for

$$x = [1, 10, 100, 1000].$$

[A] / [CO3]

13. Write a Python program to plot a circle of radius $a = 5$ using parametric equations

$$x = a\cos\theta, \qquad y = a\sin\theta,$$

where $0 \leq \theta \leq 2\pi$. Label the axes and display the resulting curve.

[A] / [CO4]

14. Write a Python program using `meshgrid()` to generate the surface

$$z = \sin(x) + \sin(y),$$

for $-3\pi \leq x, y \leq 3\pi$. Use a suitable 3D plotting method to display the surface and explain the role of the meshgrid. **[S] / [CO4]**

Part C
(Essay Questions)
Answer any 2 questions
Each question carries 10 marks

15. Write a Python program to process the following list:

$$L = [123,\ 12.4,\ \text{'haha'},\ 3.4].$$

(a) Use a `for` loop to display all the elements of the list.

(b) Insert a new element after 12.4 and append another element to the list.

(c) Explain the difference between inserting and appending an element to a Python list.

[A] / [CO2]

16. Write a Python program using NumPy to create a one-dimensional array containing the elements

$$5,\ 10,\ 15,\ 20,\ 25.$$

Using the same array, perform the following operations:

(a) Display the array.

(b) Extract the first three elements using slicing.

(c) Find the number of elements in the array.

(d) Create a new array containing the elements multiplied by 2.

[A] / [CO3]

17. Given the two vectors

$$A = [1, 2, 3], \qquad B = [4, 5, 6],$$

write a Python program using NumPy to:

(a) Calculate the dot and cross product of A and B.

(b) Display the results.

(c) Explain the difference between dot product and cross product.

[A] / [CO3]

18. Write a Python program using Matplotlib to plot the function

$$z = \sin(x) + \sin(y)$$

as a three-dimensional surface for

$$-3\pi \leq x, y \leq 3\pi.$$

Use `meshgrid()` to generate the coordinate matrices and `plot_surface()` to display the surface. Explain the steps involved and describe the role of the meshgrid in the program. **[S] / [CO4]**